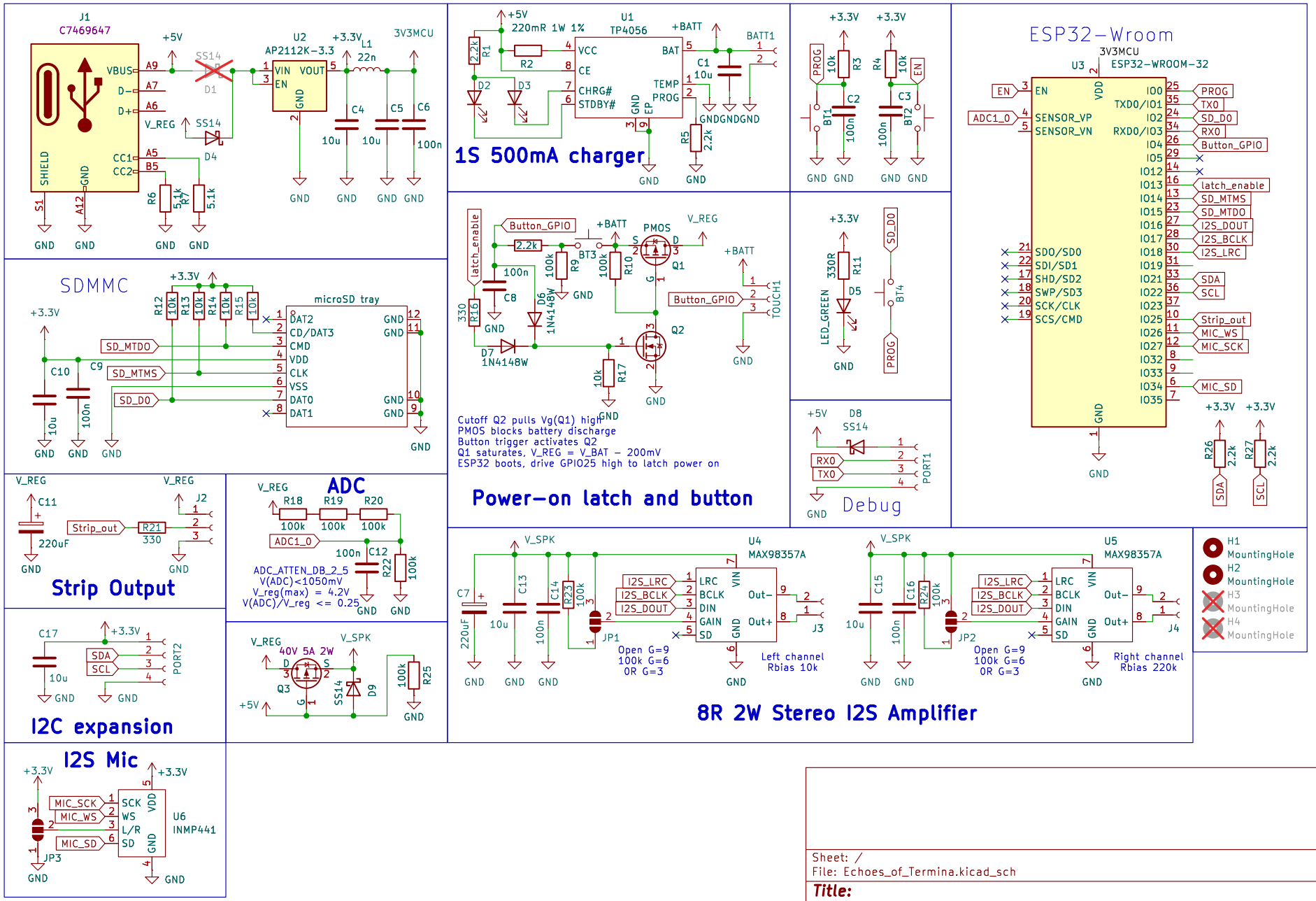

J1
C7469647
VBUS
D-
D+
CC1
CC2
SHIELD
GND
A9
A7
A6
A5
B5
A12
S1
+5V
V_REG
SS14
D1
D4
R6
5.1k
R7
5.1k
U2
AP2112K-3.3
VIN
EN
VOUT
GND
+3.3V
L1
22n
3V3MCU
C4
10u
C5
10u
C6
100n
GND
U1
TP4056
+5V
220mR 1W 1%
R1
2.2k
R2
VCC
CE
CHRG#
STDBY#
BAT
TEMP
PROG
EP
+BATT
BATT1
C1
10u
D2
D3
R5
2.2k
1S 500mA charger
PROG
10k
R3
R4
EN
C2
100n
C3
100n
BT1
BT2
ESP32-Wroom
U3
ESP32-WROOM-32
EN
ADC1_0
SENSOR_VP
SENSOR_VN
SDO/SD0
SDI/SD1
SHD/SD2
SWP/SD3
SCK/CLK
SCS/CMD
VDD
IO0
TXD0/IO1
IO2
RXD0/IO3
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
PROG
TX0
SD_D0
RX0
Button_GPIO
latch_enable
SD_MTMS
SD_MTDO
I2S_DOUT
I2S_BCLK
I2S_LRC
SDA
SCL
Strip_out
MIC_WS
MIC_SCK
MIC_SD
R26
2.2k
R27
2.2k
SDMMC
microSD tray
SD_MTDO
SD_MTMS
SD_D0
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
R12
10k
R13
10k
R14
10k
R15
10k
C10
10u
C9
100n
Power-on latch and button
Button_GPIO
latch_enable
2.2k
100n
C8
R9
100k
BT3
R10
100k
PMOS
Q1
Q2
R16
330
D6
1N4148W
D7
1N4148W
R17
10k
TOUCH1
Cutoff Q2 pulls Vg(Q1) high
PMOS blocks battery discharge
Button trigger activates Q2
Q1 saturates, V_REG = V_BAT - 200mV
ESP32 boots, drive GPIO25 high to latch power on
LED_GREEN
330R
R11
D5
BT4
Debug
D8
SS14
RX0
TX0
PORT1
Strip Output
C11
220uF
Strip_out
R21
330
J2
ADC
R18
100k
R19
100k
R20
100k
ADC1_0
C12
100n
R22
100k
ADC_ATTEN_DB_2_5
V(ADC)<1050mV
V_reg(max) = 4.2V
V(ADC)/V_reg <= 0.25
I2C expansion
C17
10u
SDA
SCL
PORT2
Q3
40V 5A 2W
SS14
D9
R25
100k
I2S Mic
MIC_SCK
MIC_WS
MIC_SD
SCK
WS
L/R
SD
VDD
GND
U6
INMP441
JP3
8R 2W Stereo I2S Amplifier
V_SPK
U4
MAX98357A
U5
MAX98357A
C7
220uF
C13
10u
C14
100n
R23
100k
JP1
Open G=9
100k G=6
OR G=3
Left channel
Rbias 10k
J3
C15
10u
C16
100n
R24
100k
JP2
Right channel
Rbias 220k
J4
LRC
BCLK
DIN
GAIN
SD
Out-
Out+
VIN
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
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